Ruby on Rails

Chances are, you already know that Rails is a framework for developing database-backed web applications in Ruby. Rails is so popular that many people get into Ruby just so that they can use Rails.

Even if you don't know much about Rails and its features, you can still follow along on this tour. We'll focus on the Rails source code, not on the features. Whenever features *are* important to understand the source code, I'll take the time to demonstrate them. Although you don't have to, you might want to get a quick introduction to Rails on its official site[1] if you're completely new to it.

While touring the Rails source code, I'll show you the snippets of code I want to focus on. However, you might also want to keep the source code handy to explore it on your own. To do that, you need to install Rails.

Installing Rails

Because Rails is always evolving, it's quite possible that the source code will have changed significantly by the time you read this chapter. Luckily, you can easily install the same version of Rails that I used to write this book, by typing `gem install rails -v 4.1.0`.

Some of the next few chapters also discuss code from a much older version of Rails, to show you how Rails' source code has evolved over time. If you wish, you can install this older version alongside the more recent one, by typing `gem install rails -v 2.3.2`.

Running the commands above installs all the gems that make up Rails 4.1.0 and 2.3.2. The `rails` gem just contains helpers, such as code generators and Rake tasks, as well as the glue code that binds together the other gems. Those other gems are the ones that do the real work. Three of the most important ones are `activerecord` (which maps application objects to database tables), `actionpack` (which deals with the "web" part of the web framework), and `activesupport` (which contains utilities for generic problems, such as time calculations and logging).

The Rails Source Code

When referring to a specific source file, I'll give you the file's path inside the system's `gems` directory, such as `gems/activerecord-4.1.0/lib/active_record.rb`. If you want to explore on your own, you can use RubyGem's `unpack` command to

1. http://rubyonrails.org